

RECOMMENDED TOLERANCE: 0.05mm

Helpful Sources:

<https://blog.accupathmed.com/precision-drawn-nitinol-tube-selection-tips-tight-tolerances/>

<https://www.mdpi.com/1424-8220/23/7/3709>

- For Nylon:
- The tubes are assembled into a straight metal tube. Diamond-shaped cutouts are machined in the walls of the inner and outer tubes to reduce the stiffness. The outer tube's diamond-shaped cutout is twice the size of the inner tube to balance the stiffness of the tubes. The difference between the outer tube's inner diameter and the inner tube's outer diameter is 0.4 mm. Kinematic experiments conducted below demonstrate that a clearance of less than 0.5 mm significantly affects the end path of the concentric tube robots.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11189652>

- recent numerical study indicated that the clearance could contribute up to 38.11% tip error when a miniature continuum robot deployed within a trocar that has a clearance of 0.25mm

Problem:

The Nylon tubing given in the open source CTCR is a different size than commercially available stock tubing for Nitinol tubes.

Solution:

- Match nitinol size difference/tolerances to nylon size difference/tolerances
 - Inner tube OD -> Middle tube ID
 - Inner tube OD == 1.9812mm
 - Middle tube ID == 2.1336mm
 - Clearance == 0.1524mm
 - Middle tube OD -> Outer tube ID
 - Outer tube ID == 2.794mm
 - Middle tube OD == 2.6416mm
 - Clearance == 0.1524mm
 - Differences are the same
 - Diameter Tolerance (Nylon): 0.0508mm
 - Exactly 33% of tube clearance, appears intentional
- Keep wall thickness the same for consistency in heat treat/flexibility/durability?
 - Need to consider potential changes between materials...
- Nitinol Equivalent
 - Nitinol standard manufacturing tolerance: 0.005 to 0.01 mm
 - Proposed clearance: 0.05mm - 0.1mm?
 - Inner Tube
 - https://chamfr.com/product/superelastic-nitinol-tubing-0-0398-od-0-026-id-39-37-length-quantity-1-bag-by-resonetics/?source_id=1775621218

- OD: 1.01092
- Middle Tube ID $\rightarrow$ OD + 0.05 == 1.06092
- Middle Tube
 - https://chamfr.com/product/nitinol-tubing-0-042-id-0-05-od-43-length-superelastic-standard-grade-overrun-quantity-1-bag-by-resonetics/?source_id=1775621423
 - ID: 1.0668mm (Clearance: 0.05588mm)
 - OD: 1.27mm
 - Outer Tube ID $\rightarrow$ OD + 0.05 == 1.32mm
- Outer Tube
 - https://chamfr.com/product/nitinol-tubing-0-052-id-0-0604-od-74-length-superelastic-standard-grade-overrun-quantity-1-bag-by-resonetics/?source_id=1775621717
 - ID: 1.3208mm (Clearance: 0.0508mm)
 - OD: 1.534mm